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Light emitting diode fixtures for vehicle photographic systems

Abstract

A light fixture is provided for use in photographic chambers that provides a combination of high flux, the ability to place/tuck the light fixture against a wall, and the ability to flash/tune the light fixture dynamically. The light fixture is based on light emitting diode elements to provide an energy efficient light source that generates far less heat than traditional mercury arc based lamps. The light fixture is modular and can be daisy chained to supply power and/or control signals to multiple attached light fixtures. Individual light fixtures may be joined together to form an array of light fixtures. The light fixtures may utilize the digital multiplex DMX512 standard for digital communication networks for controlling the light elements within the fixtures. The light fixture has a light output that is capable of providing a wide beam angle.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of, and claims priority to, U.S. application Ser. No. 17/635,746, entitled “Light Emitting Diode Fixtures For Vehicle Photographic Systems,” filed Feb. 16, 2022. U.S. application Ser. No. 17/635,746 is a National Stage Entry of PCT/US2020/043966, entitled “Light Emitting Diode Fixtures For Vehicle Photographic Systems,” filed Jul. 29, 2020. PCT/US2020/043966 claims the priority benefit of U.S. Provisional Application Ser. No. 62/880,313, entitled “Light Emitting Diode Fixtures For Vehicle Photographic Systems,” filed Jul. 30, 2019. The contents of these applications are hereby incorporated by reference in their entirety for all purposes.

FIELD OF THE INVENTION

(1) The present invention in general relates to a system for photographing vehicles; and in particular, to a modular light fixture for use in photographic enclosures for vehicle photography.

BACKGROUND OF THE INVENTION

(2) Online auto sales and auto auctions have been growing in popularity. One of the most popular online auctions to buy vehicles from is eBay™. On eBay Motors™, a user can create an account and put their vehicles up for auction. Other popular websites include Cars.com™ Typically, online vehicle sales are based solely on images of the vehicle, since the buyer is in a remote location and is unable to view the vehicle in question in person. Thus, many high quality images are required of the vehicle from many perspectives to allow a buyer to gain an understanding of a subject vehicle's condition and appearance. FIGS. 1A and 1B are line drawings of typical non-studio quality vehicle photographs that are used to list vehicles for sale. The images shown in FIGS. 1A and 1B are of low quality and require a dedicated employee to move vehicles and physically move around the subject vehicle to take the pictures. FIG. 1C illustrates reflection of the floor on the surfaces of the vehicle. The photographer also has to manually collate the images, name, and upload the files, which is a labor intensive and time consuming task, considering one photographer may be required to shoot upwards of 75 vehicles a day.

(3) Furthermore, producing high quality images is not only time consuming, but is costly and requires a studio set up. Vehicle images are particularly hard to obtain without unwanted reflections of the photographer or the surroundings; however reflection free images are critical to be able to discern surface imperfections, scratches, and dents on a vehicle surface. FIGS. 2A and 2B illustrate an existing studio configuration **1** for generating a contrast break fine **2** on a vehicle with additive lighting. The bottom edge **4** of a light box **6** creates the break fine **2** between highlight and shadow. FIGS. 3A and 3B illustrate an existing studio configuration **8** for generating a contrast break line **2** on a vehicle with subtractive lighting. The bounce fill light is a large source and makes a soft reflection in the sheet metal. A gray wall is added to “subtract” the reflection from the lower half of the truck to create contrast and shape.

(4) While these studio shots are effective in creating high quality vehicle images, the studio shots are not amenable to the high throughput required for high volume vehicle sales. Thus there is a need to be able to rapidly produce high quality reflection free images of vehicles from multiple angles and perspectives.

(5) In order to increase throughput for creating high quality vehicle images a drive through photographic tunnel as disclosed in U.S. Pat. No. 10,063,758 issued Aug. 28, 2018 and a circular dome photographic booth as disclosed in U.S. patent application Ser. No. 15/834,374 filed Dec. 7, 2017 both of which are included herein in their entirety have been implemented. Both the photographic tunnel and circular dome have a plurality of cameras mounted within the walls for recording images of a vehicle that are uploaded to a database that is used to generate image data from the vehicle positioned in the structure. The systems further include a lighting system and a tracking system to actuate one or more of the plurality of cameras and lights in a predetermined sequence and combination based on the position of the vehicle within the photographic structure.

(6) U.S. Pat. No. 9,046,740 discloses a vehicle photographic studio with a stationary circular

platform a vertical upright curved wall defining a backdrop that partially surrounds the platform, a front curved overhead member mounted across a side opening in the wall, and a plurality of spaced apart rollers rotatably mounted on a lower end portion of the wall and engaged with the track so as to mount the wall upon the track for undergoing revolvable movement along the track and about the platform to relocate the side opening of the wall at any selected angular position in relation to the platform.

(7) However, these existing vehicle photographic solutions utilize lighting systems that have lighting elements that do not provide a combination of high flux, the ability to place/tuck the lighting element against a wall, and the ability to flash/tune the lighting element dynamically. Existing strobe lights **10** are shown in a vehicle photographic booth in FIGS. **4A-4D**. The strobe lights **10** are shown positioned or mounted behind a gray wall **12**, and projecting light on to a wall **14**. However, the strobe lights **10** generate bright light casing mercury arc elements that are very hot and need to be well ventilated and cannot be placed against surfaces in order to avoid overheating or fire. Furthermore, mercury arc lights and other intense flux lights lack the ability to flash or tune dynamically.

(8) Thus, there continues to be need for improved lighting fixtures for use in photographic chambers that provide a combination of high flux, the ability to place/tuck the lighting element against a wall, and the ability to flash/tune the lighting element dynamically.

SUMMARY OF THE INVENTION

(9) A number of embodiments can include a light fixture. The light fixture can comprise a housing, a rotatable heatsink coupled to the housing, and one or more lighting lenses coupled to the rotatable heatsink. A property of light emitted through at least one of the lighting lenses can be changed when the rotatable heatsink is rotated relative to the housing.

(10) Many embodiments can include a method of using a light fixture. The method can comprise coupling a rotatable heatsink to a housing, rotating the rotatable heatsink relative to the housing, and activating one or more lights emitted through one or more lighting lenses coupled to the rotatable heatsink. A property of light emitted through at least one of the lighting lenses can be changed when the rotatable heatsink is rotated relative to the housing.

(11) Various embodiments can include a method of manufacturing a light fixture. The method can comprise coupling one or more lighting lenses to a rotatable heatsink and coupling the rotatable heatsink to a housing. A property of light emitted through at least one of the lighting lenses can be changed when the rotatable heatsink is rotated relative to the housing.

(12) A modular light fixture is provided. The modular light fixture includes a housing, a heatsink secured to the housing with securements, and a lighting lens supported on the heatsink that contains an array of light elements. One or more fans are within the housing, along with a set of power and data connections.

(13) A vehicle photographic system is provided. The vehicle photographic system includes a photographic booth or structure, one or more cameras, one or more of the light fixtures as disclosed above, a computer workstation, and a controller for the one or more cameras and the one or more light fixtures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter that is regarded as the invention is particularly pointed out and distinctly claimed in the claims at the conclusion of the specification. The foregoing and other objects, features, and advantages of the invention are apparent from the following detailed description taken in conjunction with the accompanying drawings in which:

(2) FIGS. **1A** and **1B** are typical non-studio quality vehicle photographs rendered as line drawings;

- (3) FIG. 1C illustrates a photograph of a studio shot with uncontrolled reflections in a vehicle surface;
- (4) FIGS. 2A and 2B illustrate an existing studio configuration for generating a contrast break line photographs on a vehicle with additive lighting rendered as line drawings;
- (5) FIGS. 3A and 3B illustrate an existing studio configuration for generating a contrast break line photographs on a vehicle with subtractive lighting rendered as line drawings;
- (6) FIGS. 4A-4D illustrate the use of existing strobe lighting elements in a vehicle photographic booth;
- (7) FIGS. 5A-5C are a series of perspective views of a modular light fixture in accordance with embodiments of the invention;
- (8) FIG. 6 illustrates air blower nozzles for cooling and for cleaning hard to reach fans in an array of modular light fixtures in accordance with embodiments of the invention;
- (9) FIGS. 7A-7E illustrate power connections between the modular light fixtures of FIG. 5 according to an embodiment of the invention;
- (10) FIGS. 8A-8D illustrate control line connections between the modular light fixtures of FIG. 5 in accordance with embodiments of the invention;
- (11) FIGS. 9A and 9B are pictures showing the placement of the modular light fixture of FIG. 5 on the floor and against a wall of a photographic chamber, and mounted to a VESA mount, respectively, in accordance with embodiments of the invention;
- (12) FIGS. 10A-10C illustrate the rotation of the lens assembly to change the spread of the light from the modular light fixture in accordance with embodiments of the invention;
- (13) FIGS. 10D and 10E illustrate a naturally occurring sundog and a sundog generated in a photographic dome structure with an embodiment of the modular light fixture of FIG. 5, respectively;
- (14) FIGS. 11A and 11B illustrate arrays and signal connections formed there between with the modular light fixtures of FIG. 5 in accordance with embodiments of the invention; and
- (15) FIG. 12 is a system block diagram of the major components for photographing vehicles in accordance with embodiments of the invention.

DESCRIPTION OF THE INVENTION

- (16) The present invention has utility as a modular lighting fixture for use in photographic chambers that provides a combination of high flux, the ability to place/tuck the lighting fixture against a wall, and the ability to flash/tune the lighting fixture dynamically. Embodiments of the inventive lighting fixture may be based on light emitting diode elements to provide an energy efficient light source that generates far less heat than tradition mercury are based lamps. Embodiments of the inventive lighting fixture may be modular and may be daisy chained to supply power and/or control signals to multiple attached lighting fixtures. In a specific inventive embodiment individual lighting fixtures may be joined together to form an array of lighting fixtures. Specific inventive embodiments of the lighting fixtures may utilize the digital multiplex DMX512 standard for digital communication networks for controlling the lighting elements within the fixtures. In a specific inventive embodiment of the lighting fixture, light output is rated at 300 Watts (94 CRI) and is capable of providing a wide beam angle.
- (17) In addition to the use of inventive embodiments of the modular lighting fixture in photographic chambers, the modular lighting fixtures are also very good lights for lighting theatrical backgrounds and green screens. The high output and low profile of the modular lighting fixture allows the lighting fixtures to be easily hidden. Furthermore, the extra wide lenses allow that bright lights to be evenly spread and provide exceptional blending with a neighboring fixture.
- (18) It is to be understood that in instances where a range of values are provided herein, that the range is intended to encompass not only the end point values of the range, but also intermediate values of the range as explicitly being included within the range and varying by the last significant figure of the range. By way of example, a recited range of from 1 to 4 is intended to include 1-2, 1-

3, 2-4, 3-4, and 1-4.

(19) Referring now to the figures, FIGS. 5A-5C are a series of perspective views of an inventive embodiment of a modular light fixture **20**. The modular light fixture **20** has a housing **22** that is secured to a heatsink **24** with securements **26**. The heatsink **24** provides a surface that supports an array of lighting elements in a lighting lens **28**. The heatsinks **24** spreads and dissipates heat generated by the lighting elements. One or more fans **30** provide cooling within the housing **22**. Electrical connections **32** supply current to a transformer or power supply (not shown) within the housing **22**. Control signal connections **34** supply signals for controlling the switching (on/off) and intensity and dimming of the array of lighting elements in a lighting lens **28**. FIG. 5C is a top view of the lighting lens **28** that more clearly shows the array of individual lighting elements **29**, which as previously noted may be light emitting diodes (LED) held in the lighting lens **28**. The individual lighting elements **29** may be evenly spaced in a series of rows that form the array in the lighting lens **28**. Different lighting lenses **28** may be used to create varying lighting effects. In a specific embodiment the color of the individual LED s may be controlled.

(20) FIG. 6 is a perspective view of an airline **36** and nozzles **38** that may be used for self cleaning in an array installation of lighting fixtures **20** as shown in FIGS. 11A and 11B. In an array installation the fans **30** of each lighting fixture **20** may be difficult to access for cleaning, and the nozzles **38** may provide a blast of air into a fan **30** on a periodic basis, for example hourly, to clear any dust or debris that may have accumulated in the vicinity of the fan **30**. In addition to self-cleaning in specific inventive embodiments, a continuous low volume of air may be supplied to keep the housing **22** of the lighting fixture **20** extra cool.

(21) FIGS. 7A-7E illustrate power connections between the modular light fixtures **20**. Inventive embodiments may use daisy chain power connections for simplified and quick installations. In a specific inventive embodiment using a 220 volt alternating current (VAC) power supply with a 20 amp rating up to sixteen lighting fixtures may be daisy chained or connected together. In 220 VAC operation a National Electrical Manufacturers Association (NEMA) 6-20P to International Electrotechnical Commission (IEC) C13 power cord **40**, as shown in FIG. 7A, may be used to supply power to the first lighting fixture **20** in a daisy chain configuration, an example of which is shown in FIG. 7C. In a specific inventive embodiment using a 115 volt alternating current (VAC) power supply with a 15 amp rating up to six lighting fixtures may be daisy chained or connected together. In 115 VAC operation a NEMA 5-15P to IEC C13 power cord **42**, as shown in FIG. 7B, may be used to supply power to the first lighting fixture **20** in a daisy chain configuration, an example of which is shown in FIG. 7C. In a specific inventive embodiment, each lighting fixture **20** draws a maximum of 2.5 amps at 115 VAC. In either 115 or 220 VAC operating configurations, an electrical cord, such as an IEC C14 to IEC C13 cord, and electrical junction connectors **32**, as shown in close up in FIGS. 7D and 7E, respectively are used to establish connections between each of the lighting fixtures **20**. It is appreciated that additional electrical standards may be used for power connection and distribution.

(22) FIGS. 8A-8D illustrate control line connections between the modular lighting fixtures **20**. In specific inventive embodiments the digital multiplex DMX 512 standard is used to send control signals to the lighting fixtures **20**. FIG. 8A shows a standalone DMX 512 network controller **46** connected to a series of embodiments of the inventive lighting fixtures **20** in a daisy chain configuration. As best seen in FIGS. 8B and 8C data connection jacks **34** may be based on the RJ45 standard which is typically used with Ethernet cables. FIG. 8C shows an inventive embodiment of the lighting fixture **20'** with a built in DMX 512 decoder or manual standalone controller **48**. FIG. 8D is a close-up view of the decoder/controller **48**. It is appreciated that additional signal control standards may be used for power connection and distribution.

(23) FIGS. 9A and 9B are pictures showing the placement of the modular light fixture **20** on the floor **50** and against a wall **14** of a photographic chamber, and mounted to an adjustable arm mount that may be compatible with a Video Electronics Standards Association (VESA) mount,

respectively. The lay flat design of the modular light fixture **20** allows for minimal or no additional mounting hardware or brackets when resting on a flat surface. The lay flat/set-down design makes installing the modular light fixture **20** into a photographic booth or car stage fast and efficient with minimal labor with no additional hardware needed. As seen in FIG. 5B, a modular light fixture **20** may also be placed in a standing position with no additional hardware required.

(24) FIGS. **10A-10C** illustrate the rotation of the lens assembly **28** to change the spread of the light from the modular light fixture. In FIG. **10A** securements **26** are removed to detach the heatsink **24** with the mounted lighting lens **28**. As further shown in FIG. **10A**, the heatsink **24** is rotated 180 degrees to change the spread of the light. FIGS. **10B** and **10C** illustrate the differences in the light spread between the rotated lighting lens. The rotation of the heatsink **24** result in the flipping (rotation) of the lighting lens **28** which provides two lighting effects per each available lighting lens. FIG. **10D** is a photograph of a naturally occurring sundog. A sundog is a concentrated patch of sunlight occasionally seen about 22° to the left or right of the sun. Sundogs often form pairs on either side of the sun when sunlight refracts through icy clouds containing hexagonal plate crystals aligned with their large, flat faces parallel to the ground. In a specific embodiment the modular light fixture (**20**, **20'**) may be arraigned to mimic a sundog in a domed structure as shown in FIG. **10E** where the light is cast on the wall **14** of a photographic chamber from behind the gray wall **12**.

(25) FIGS. **11A** and **11B** illustrate arrays and signal connections formed therebetween with the modular light fixtures **20**. FIG. **11A** shows an array **60** formed with six modular light fixtures **20**. The overhang of the heatsink **24** beyond the housing **22** allows for an offset area for cabling to be run between the light fixtures **20** in the array. The offset area provides for cable management of power and control lines between the individual lighting fixtures **20**. FIG. **11B** shows a signal bus **62** running between the light fixtures **20** in a three unit array **70**.

(26) FIG. **12** is a system block diagram **100** of the major components for photographing vehicles in accordance with embodiments of the invention. A central process unit (CPU) **122** coordinates and controls the overall operation of the photographing system **100** that may be operated in a photographic structure or booth. The communication interface **124** is representative of various links and input connections and devices illustratively including but not limited to wired and wireless links, optical drives, universal serial bus (USB), flash memory slot, and combinations thereof, for receiving and sending data that may or may not be in real time. The bus **102** links the various components in the system. Memory **116** serves as storage for operating programs and firmware for the photographic system **100**. A database with vehicle and client information is stored in Memory **116**. Memory **116** is made up of read-only memory ROM and random access memory (RAM).

(27) Graphics chipset **120** drives a display. The display may be liquid crystal display (LCD), light emitting diode (LED), or other known display technologies. Control interface **106** may include knobs, buttons, and other touch sensitive controls for selecting various menu items or for inputting alphanumeric information. Sensors **104** sense the presence of a vehicle and the vehicle position. RFID/barcode **108** is a reader that detects and interprets tags mounted to vehicles or driver identification (ID) cards. Cameras **110** and lighting **114**, which include embodiments of the modular light fixture **20**, are controlled and sequenced by the CPU **122**. The CPU may utilize the DMX 512 standard to control the lighting fixtures **20**. Indicators **112** provide visual feedback to system users. In inventive embodiments, the central process unit (CPU) **122**, or master and node computers, in the case of many cameras in use—these processing nodes speedup download and system throughput, and the CPU **122** coordinates and controls the plurality of cameras in the photographic structure. In specific inventive embodiments, a communication interface connects the plurality of cameras via one or more of wired and wireless links, optical drives, universal serial bus (USB), flash memory slot, and combinations thereof for receiving and sending data.

(28) The use of sequenced and automated image capture allows for the rapid image processing of vehicles for auction sales, dealer records, as well as car condition assessments for rental agencies, fleet management companies, public safety agencies, municipal and government agencies, etc. A

complete set of vehicle images are collected in between 5 and 90 seconds allowing for high throughput imaging associated with an auto auction or manufacturer.

(29) Embodiments of the inventive enclosable photographic booth allow for a complete set of multi perspective high quality vehicle images to be obtained and recorded into a template or placed in a shared network folder in less than 90 seconds, and in some instances in approximately 5, 10, or 15 seconds. Thus, once a subject vehicle is photographed in the inventive enclosable oblong hemispherical domed photographic booth, the vehicle is available and ready for sale online based on the uploaded images and vehicle identification number (VIN) uploaded information. The rapid image processing and recordation of vehicle condition allows for new business models such as creating virtual or cyber dealerships where a wholesale customer never takes possession of a subject vehicle, and the vehicle is dropped shipped to the end retail customer. Photos obtained with the inventive system may be sold along with the subject vehicle for use by a purchasing used car retailer, and as it typically takes four days for the buyer to take delivery of the wholesale vehicle, with instant access to the vehicle photos the purchaser can start advertising the vehicle typically four days prior to physically taking possession of the vehicle.

(30) Embodiments of the inventive photographic booth may utilize radio frequency identification (RFID) tags to identify and record vehicles as the vehicles are processed through the system. The RFID may be related to the vehicle identification number (VIN) of a vehicle to be processed. It is further appreciated that additional identifiers may be used illustratively including barcodes that relate to the VIN. Information related to the vehicle, such as vehicle make, model, body style, and color encoded in the VIN may be used to automatically adjust lighting. Further information that may be related to the RFID or barcode may include a dealership setting that may also indicate which dealer the vehicle is being shot for, and incorporate that dealer's specific preferences such as lighting style, file size and format, number of photo shots and angles to be recorded. RFID or barcode information, read manually or automatically may also be used to project text onto the background of the shot or overlay text onto the file during processing, such as price, dealer name, vehicle specifications, mileage, etc.

(31) As a person skilled in the art will recognize from the previous detailed description and from the figures and claims, modifications and changes can be made to the preferred embodiments of the invention without departing from the scope of this invention defined in the following claims.

Claims

1. A light fixture comprising: a housing; a rotatable heatsink coupled to the housing; wherein the rotatable heatsink comprises an overhang extending beyond an edge of the housing, and wherein the overhang creates an offset allowing a cable to pass between the light fixture and another light fixture arranged in an array; and one or more lighting lenses coupled to the rotatable heatsink, wherein a property of light emitted through at least one of the one or more lighting lenses is changed, in response to the rotatable heatsink being rotated relative to the housing.
2. The light fixture of claim 1, further comprising an airline with a set of nozzles that supply air in periodic bursts to clean components of the light fixture.
3. The light fixture of claim 1, further comprising an airline with a set of nozzles that supply a continuous supply of air to cool components of the light fixture.
4. The light fixture of claim 1, further comprising a signal bus configured to join a plurality of light fixtures in the array.
5. The light fixture of claim 1, wherein the property of the light comprises a light spread of the one or more lighting lenses.
6. The light fixture of claim 1, wherein the rotatable heatsink is coupled to a top portion of the housing.
7. The light fixture of claim 1, wherein, in response to the rotatable heatsink being rotated, a

sundog light formation is emitted by the light fixture.

8. A method of using a light fixture, the method comprising coupling a rotatable heatsink to a housing; rotating the rotatable heatsink relative to the housing to change a property of light emitted through at least one of the one or more lighting lenses; arranging the light fixture and another light fixture in an array; passing a cable between the light fixture and the another light fixture and through an offset created by an overhang of the rotatable heatsink, wherein the overhang extends beyond an edge of the housing; and activating one or more lights emitted through one or more lighting lenses coupled to the rotatable heatsink.

9. The method of claim 8, further comprising activating an airline with a set of nozzles that supply air in periodic bursts to clean components of the light fixture.

10. The method of claim 8, further comprising activating an airline with a set of nozzles that supply a continuous supply of air to cool components of the light fixture.

11. The method of claim 8, further comprising joining a plurality of light fixtures in the array using a signal bus.

12. The method of claim 8, wherein the property of the light comprises a light spread of the one or more lighting lenses.

13. The method of claim 8, wherein the coupling the rotatable heatsink comprises coupling the rotatable heatsink to a top portion of the housing.

14. The method of claim 8, further comprising rotating the rotatable heatsink to create a sundog light formation.

15. A method of manufacturing a light fixture, the method comprising: coupling one or more lighting lenses to a rotatable heatsink; coupling the rotatable heatsink to a housing, wherein a property of light emitted through at least one of the one or more lighting lenses is changed, in response to the rotatable heatsink being rotated relative to the housing; arranging the light fixture and another light fixture in an array; and passing a cable between the light fixture and the another light fixture through an offset created by an overhang of the rotatable heatsink, wherein the overhang extends beyond an edge of the housing.

16. The method of claim 15, further comprising placing an airline with a set of nozzles that supply air in periodic bursts to clean components of the light fixture.

17. The method of claim 15, further comprising placing an airline with a set of nozzles that supply a continuous supply of air to cool components of the light fixture.

18. The method of claim 15, further comprising joining a plurality of light fixtures in the array using a signal bus.

19. The method of claim 15, wherein the property of the light comprises a light spread of the one or more lighting lenses.

20. The method of claim 15, wherein, in response to the rotatable heatsink being rotated, a sundog light formation is emitted by the light fixture.
